

Implement a file copy utility using low-level Linux file I/O system calls (open, read, write, lseek, close). Compare its performance with standard library functions (fopen, fread, fwrite).

Develop a program that redirects standard input and output using file descriptors and dup2(). Demonstrate how shells implement I/O redirection.

```
[varnikakomali@VARNIKAs-MacBook-Air Practical % nano practical9.c
[varnikakomali@VARNIKAs-MacBook-Air Practical %
[varnikakomali@VARNIKAs-MacBook-Air Practical % gcc practical9.c -o practical9 && ./practical9
=== Part 1: Low-Level Syscalls vs Standard Library ===
Low-Level Syscalls (open/read/write/lseek): 0.016461 seconds
Standard Library (fopen/fread/fwrite):      0.014049 seconds

=== Part 2: Shell I/O Redirection using dup2() ===
Stdout restored to terminal.
Stdin captured from file: "This message was redirected to redirect_out.txt via dup2()!"
varnikakomali@VARNIKAs-MacBook-Air Practical %
```